

JEE Main - 11 | JEE 202

Date: 25/07/2021

Maximum

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

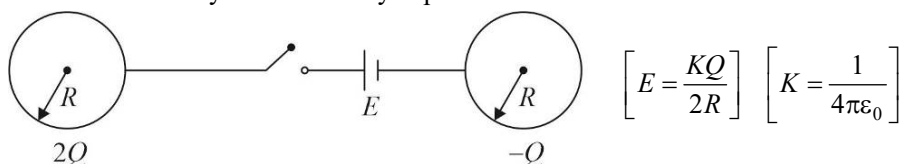
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Two large non-conducting planes carrying charge with uniform charge density σ each are placed along $x = 2$ and $y = 2$ respectively. Work done in moving a charge Q from $(0, 0, 0)$ to $(d, -d, d)$ is: [$d < 2$]

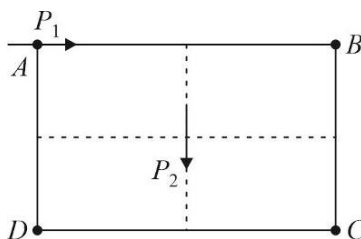
(A) $\frac{\sigma Q d}{2 \epsilon_0}$ (B) $\frac{2 \sigma Q d}{\epsilon_0}$ (C) $\frac{\sigma Q d}{\epsilon_0}$ (D) zero

2. Two conducting spheres with equal radius R having charges $2Q$ and $-Q$ are kept at a large separation. They are now connected by a wire and key is pressed as shown. Heat loss in the circuit will be:



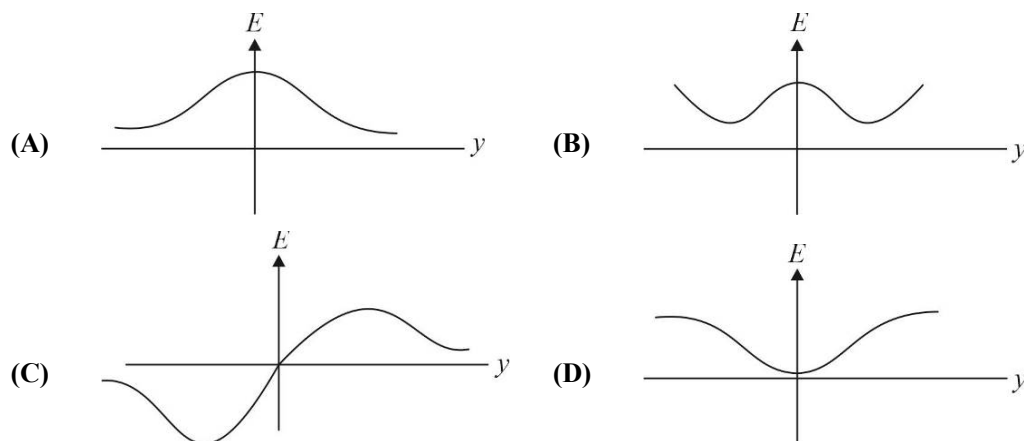
(A) $\frac{30Q^2}{64\pi\epsilon_0 R}$ (B) $\frac{16Q^2}{\pi\epsilon_0 R}$ (C) $\frac{25Q^2}{64\pi\epsilon_0 R}$ (D) $\frac{5Q^2}{8\pi\epsilon_0 R}$

3. Two small dipoles with dipole moment P_1 & P_2 are placed at the corner and center of a square as shown. Ratio of P_1 and P_2 so that net electric field at B is along BC is :

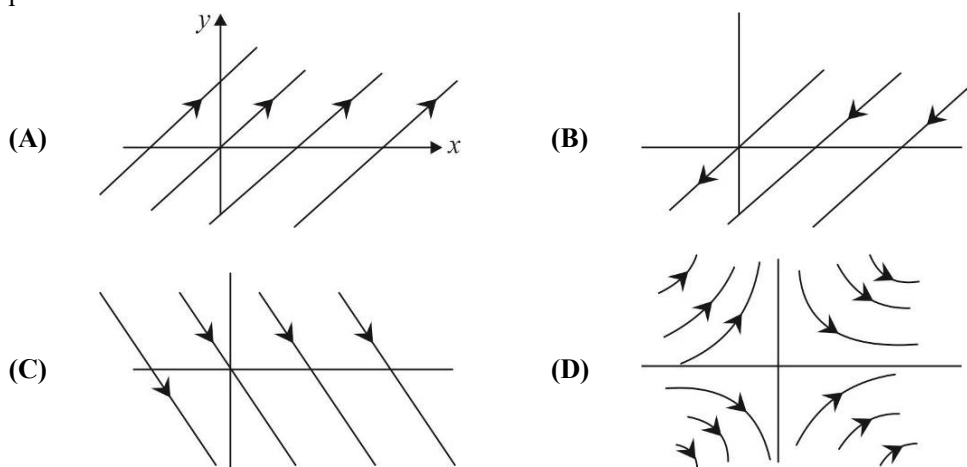


(A) $\frac{3\sqrt{2}}{2}$ (B) $2\sqrt{3}$ (C) $\frac{\sqrt{5}}{2}$ (D) $\frac{2\sqrt{7}}{3}$

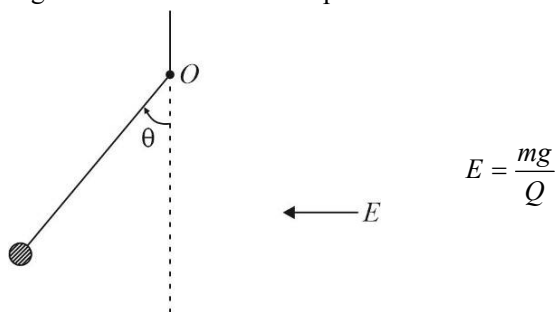
4. Two point charges Q and $-Q$ are placed on x -axis at equal distance from origin. Which of the following graphs shows correct variation of electric field at different points on y -axis with y ?



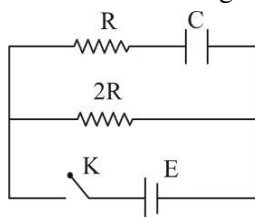
5. Electric potential in space varies according to relation $V(x, y, z) = -3x + 2y$. Correct electric field line pattern is :



6. A small bob of mass m and carrying charge Q is in equilibrium in a horizontal uniform electric field E in addition to the gravitational field g as shown. Bob is attached to point O with a string of length ℓ . Find the minimum speed to be given to the bob from this position so that it completes a vertical circle about O .

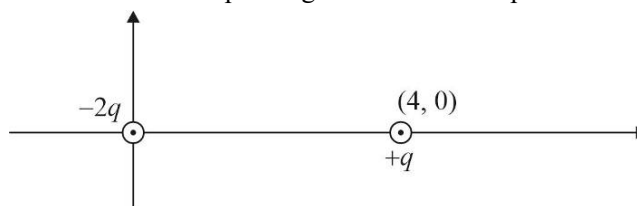


- (A) $\sqrt{5g\ell}$ (B) $\sqrt{5\sqrt{2}g\ell}$ (C) $\sqrt{10g\ell}$ (D) $\sqrt{2\sqrt{5}g\ell}$
7. In order to increase range of a voltmeter 10 folds, resistance to be connected in series with voltmeter is :
[Resistance of the voltmeter = 20Ω]
- (A) 200Ω (B) 20Ω (C) 180Ω (D) 140Ω
8. In the circuit shown key is closed at $t = 0$. Current through the battery at time $t = RC$ is:

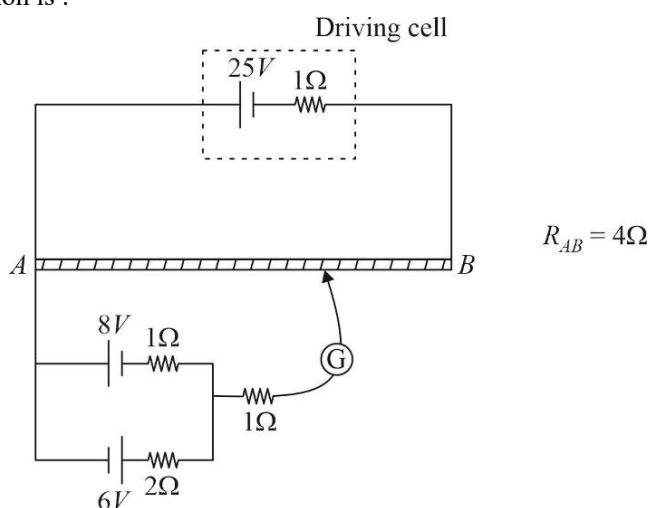


- (A) $\frac{E}{R} \left[1 + \frac{1}{e} \right]$ (B) $\frac{E}{2Re}$ (C) $\frac{E}{R} \left[\frac{e+2}{2e} \right]$ (D) $\frac{E}{2R} \left[\frac{e-2}{2e} \right]$

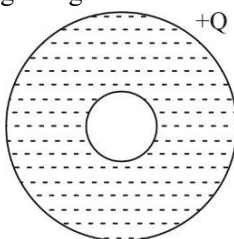
9. The shape of equipotential curve corresponding to $V = 0$ for the pair of two charges placed on x -axis is :



- (A) Ellipse
(B) Circle with centre $\left(-\frac{16}{3}, 0\right)$
(C) Circle with centre $\left(\frac{16}{3}, 0\right)$
(D) Hyperbola
10. In the potentiometer setup a driving cell with EMF $25V$ and internal resistance 1Ω is used. Length of the potentiometer wire is $150cm$ and its resistance is 4Ω . The balancing length for which galvanometer shows zero deflection is :

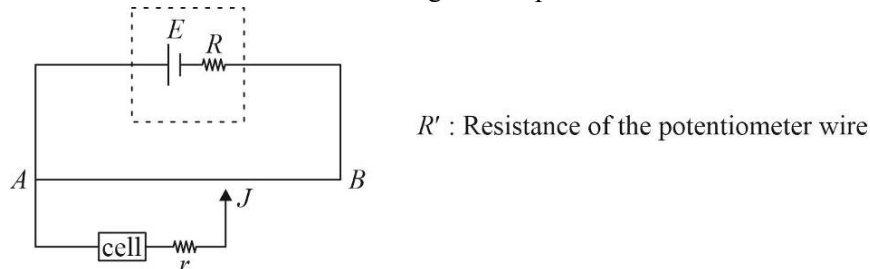


- (A) $55cm$ (B) $25cm$ (C) $45cm$ (D) $40cm$
11. A neutral conducting sphere has a spherical cavity at the center. A charge $+Q$ is given to the conductor. Choose the correct statement regarding charge on the inner surface on the cavity.

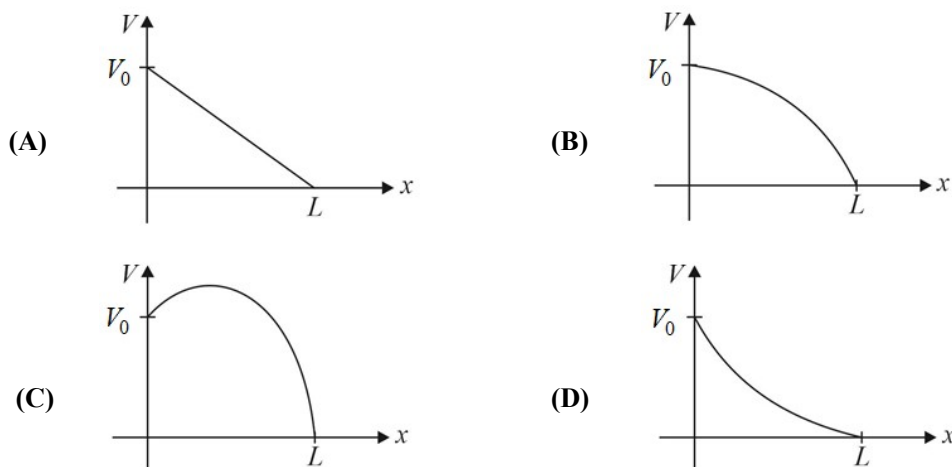
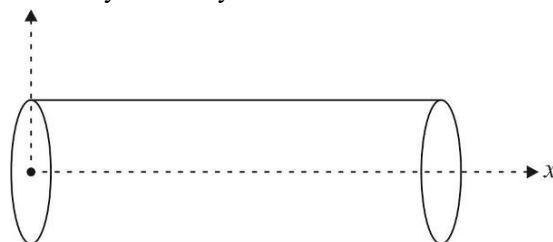


- (A) Non-zero and uniform
(B) Non-zero and non-uniform
(C) Zero
(D) Depends on the location of cavity
12. Rate of heat dissipation per unit volume in a resistance with conductivity σ and electric field E is :
- (A) $E\sigma$ (B) $\frac{E}{\sigma}$ (C) $E^2\sigma$ (D) $\frac{E^2}{\sigma}$
13. Choose the incorrect statement.
- (A) Dielectrics used in capacitors are insulators
(B) Polar dielectric have higher value of dielectric constant than non-polar dielectric
(C) Capacitance of a capacitor increases when a dielectric is inserted in it
(D) Lesser value of k (dielectric constant) means dielectric can be easily polarised

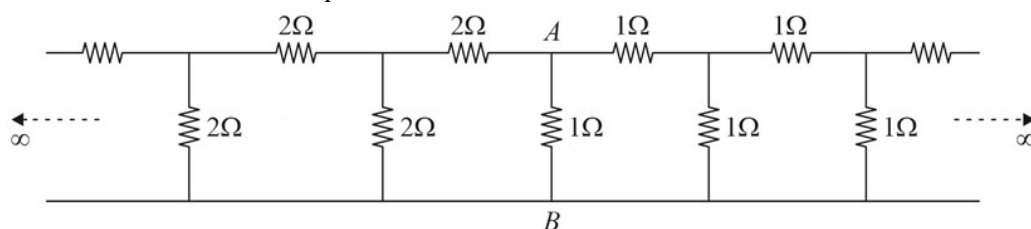
14. In the potentiometer experiment below null point was not obtained while sliding the jockey over potentiometer wire AB . Which of the following is not a possible cause for this?



- (A) Wrong terminal of cell is connected to A
 (B) Emf of the cell $\gg E$
 (C) $R \gg R'$
 (D) $r > R'$
15. Resistivity of a cylindrical wire of radius r and length L varies with x as $\rho = \rho_0 e^{-x/L}$ where x is the distance from left end. If left end of the wire is maintained at V_0 and right end at zero potential, variation of potential (V) with x is correctly shown by :

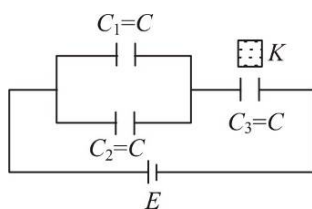


16. In the infinite ladder shown equivalent resistance across AB is :



- (A) $\frac{\sqrt{5}+1}{\sqrt{5}+4} \Omega$ (B) $\frac{\sqrt{5}+4}{\sqrt{5}+1} \Omega$ (C) $\frac{\sqrt{5}-1}{\sqrt{5}+4} \Omega$ (D) $\frac{\sqrt{5}-4}{\sqrt{5}+1} \Omega$

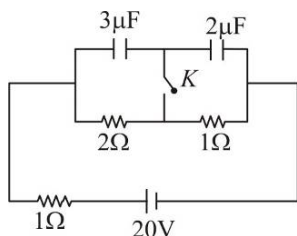
17.



In the circuit above when a dielectric slab of dielectric constant K is inserted in C_3 an extra charge $\frac{CE}{3}$ flows through the battery. Value of K is:

- (A) 2 (B) 3 (C) 4 (D) 5

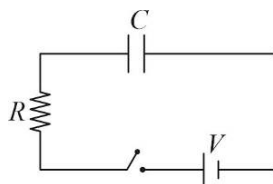
18.



In the circuit above charge on $3\mu F$ capacitor when key is open is q_1 and charge on it is q_2 when key is closed. Value of $\frac{q_1}{q_2}$ is:

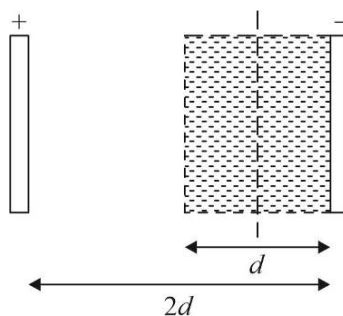
- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $\frac{3}{5}$

19. In the RC circuit shown key is closed at $t = 0$. Energy stored in the capacitor when rate of energy transfer to the capacitor is maximum is :



- (A) $\frac{1}{2}CV^2$ (B) $\frac{CV^2}{4}$ (C) $\frac{CV^2}{8}$ (D) $\frac{CV^2}{16}$

20. A capacitor with capacitance C is charged by a battery so that charge on the capacitor is Q_0 . Without removing the battery, a dielectric with dielectric constant $K = 10$ is inserted between the plates such that it occupied half the space between the plates. What is the potential difference between positive plate and centre of the dielectric?



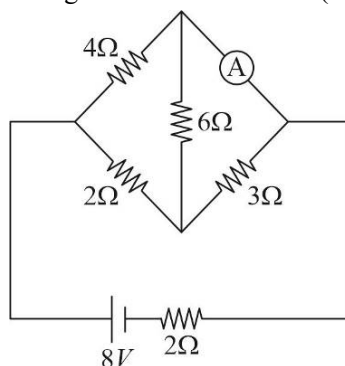
- (A) $\frac{21Q_0}{22C}$ (B) $\frac{11Q_0}{10C}$ (C) $\frac{21Q_0}{44C}$ (D) $\frac{11Q_0}{20C}$

SPACE FOR ROUGH WORK

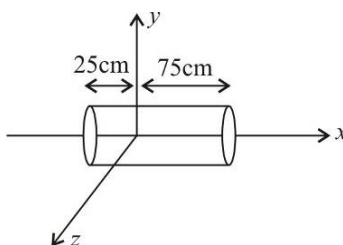
SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

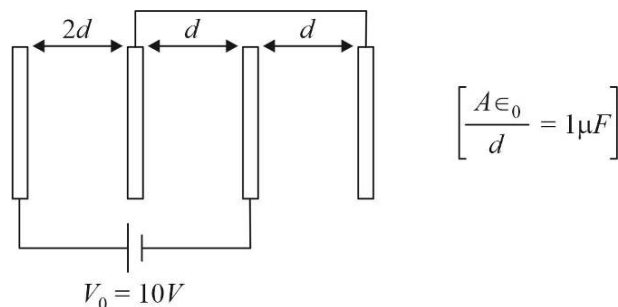
21. In the circuit shown what is the reading of the ideal ammeter (in Amperes)?



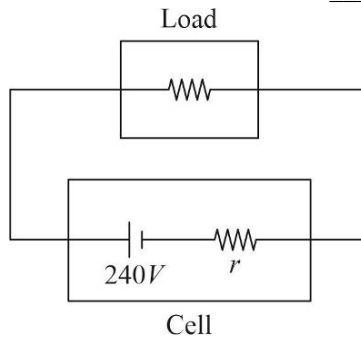
22. Electric field in a region of space is given by $\vec{E} = E_0 x \hat{i}$; where x is in meters and $E_0 = \frac{2N}{C-m}$. What is electric flux $\left(\text{in } \frac{N-m^2}{C} \right)$ through a cylindrical Gaussian surface with $R = 1m$? [Take $\pi = 3.14$]



23. Four identical thin conducting plates are arranged as shown. Plate area of each plate = A . Magnitude of charge (in μC) on plate connected to the negative terminal of the battery is _____.

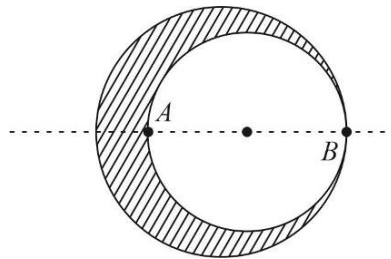


24. In the circuit shown power delivered to the load is same for load resistance equal to 9Ω and 4Ω . Maximum power (in watts) which can be delivered to load is _____.



25. A spherical portion of radius $\frac{3R}{2}$ is removed from a uniformly charged non-conducting sphere of radius $2R$ as shown. Potential difference (in volts) between diametrically opposite points A and B inside the cavity is _____.

Volume charge density on the sphere is ρ with $\frac{\rho}{\epsilon_0} = 16 \frac{N}{C-m}$ and $R = 1.5m$.



SPACE FOR ROUGH WORK

PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- A solid melt slightly above 273 K and is a poor conductor of heat and electricity. To which of the following categories does it belong?
(A) Ionic solid (B) Covalent solid (C) Metallic solid (D) Molecular solid
- In a tetragonal crystal,
(A) $\alpha = \beta = 90^\circ \neq \gamma; a = b = c$ (B) $\alpha = \beta = \gamma = 90^\circ; a = b \neq c$
(C) $\alpha = \beta = \gamma = 90^\circ; a \neq b \neq c$ (D) $\alpha = \beta = 90^\circ; \gamma = 120^\circ; a = b \neq c$
- For which of the following pair, the heat of mixing, ΔH_{mix} , is approximately zero?
(A) $\text{CH}_3\text{COOCH}_3 + \text{CHCl}_3$ (B) $\text{CH}_3\text{COOH} + \text{H}_2\text{O}$
(C) $\text{C}_2\text{H}_5\text{OH} + \text{CH}_3\text{OH}$ (D) $\text{CH}_3\text{COCH}_3 + \text{C}_6\text{H}_6$
- Liquids A and B form an ideal solution. A certain solution of A and B contains 25 mole percent of A, whereas the vapours in the equilibrium with the solution at 298 K contains 50 mole percent of A. The ratio of vapour pressures of pure A to that of pure B at 298 K, is:
(A) 1 : 1 (B) 3 : 1 (C) 1 : 3 (D) 2 : 1
- The rate expression for the reaction $\text{A(g)} + \text{B(g)} \rightarrow \text{C(g)}$ is $\text{rate} = k \cdot C_A^2 \cdot C_B^{1/2}$ where C_A and C_B are concentrations of A and B. What changes in the initial concentration of A and B will cause the rate of reaction increase by a factor of eight?
(A) $C_A \times 2, C_B \times 2$ (B) $C_A \times 2, C_B \times 4$ (C) $C_A \times 1, C_B \times 4$ (D) $C_A \times 4, C_B \times 1$
- For the elementary reaction, $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$, if the volume of a container is tripled, the rate becomes:
(A) 3 times (B) 27 times (C) $\frac{1}{27}$ times (D) as initial
- An aqueous solution of a non-volatile and non-electrolyte solute (molecular mass = 150 g mol^{-1}) boils at 373.26 K. The composition of solution, in terms of mass percent of the solute, is (K_b of water = 0.52)
(A) 50% (B) 17.5% (C) 6.98% (D) 75%
- The low density of alkali metals is due to:
(A) their BCC structure in which about 32% of the available space is unfilled
(B) their HCP structure in which about 74% of the available space is unfilled
(C) their cubic close packed (CCP) structure in which about 74% of the available space is unfilled
(D) their BCC structure in which about 47% of the available space is unfilled
- An organic compound A decomposes according to the following two parallel first-order reactions
 $\text{A} \xrightarrow{k_1} \text{B}$ and $\text{A} \xrightarrow{k_2} \text{C}$. If k_1 is $1.25 \times 10^{-5} \text{ s}^{-1}$ and $\frac{k_1}{k_2} = \frac{1}{9}$, then the value of $\frac{[\text{C}]}{[\text{B}]}$ after one hour of start of reaction by initially taking A only, is ($\ln 1.568 = 0.45$)
(A) $\frac{1}{9}$ (B) 0.5112 (C) 1.4112 (D) 9

10. The nucleus resulting from ${}_{92}^{238}\text{U}$ after successive emission of two α and four β particles is:
 (A) ${}_{90}^{230}\text{Th}$ (B) ${}_{94}^{230}\text{Pu}$ (C) ${}_{88}^{230}\text{Ra}$ (D) ${}_{92}^{230}\text{U}$
11. How many grams of sucrose (molecular weight = 342 gmol^{-1}) should be dissolved in 100 g water in order to produce a solution with a 104.76°C difference between the freezing point and the boiling point temperature? ($K_f = 1.86\text{ K Kg mol}^{-1}$, $K_b = 0.52\text{ K Kg mol}^{-1}$)
 (A) 34.2 g (B) 68.4 g (C) 684 g (D) 313.06 g
12. The ionic radii of Rb^+ and I^- are 1.46 and $2.16\text{ \AA}$, respectively. The most probable type of structure exhibited by it is:
 (A) CsCl type (B) NaCl type (C) ZnS type (D) CaF_2 type
13. Addition of arsenic in small amount to pure germanium will result in the formation of
 (A) n-type semiconductor (B) germanium arsenide
 (C) p-type semiconductor (D) a superconducting alloy
14. Liquids A and B form an ideal solution. The plot of $\frac{1}{X_A}$ (Y-axis) versus $\frac{1}{Y_A}$ (X-axis) (where X_A and Y_A are the mole fractions of A in liquid and vapour phases at equilibrium, respectively) is linear whose slope and intercept, respectively, are given as
 (A) $\frac{P_A^\circ}{P_B^\circ}, \frac{(P_A^\circ - P_B^\circ)}{P_B^\circ}$ (B) $\frac{P_A^\circ}{P_B^\circ}, \frac{(P_B^\circ - P_A^\circ)}{P_B^\circ}$
 (C) $\frac{P_B^\circ}{P_A^\circ}, \frac{(P_A^\circ - P_B^\circ)}{P_B^\circ}$ (D) $\frac{P_B^\circ}{P_A^\circ}, \frac{(P_B^\circ - P_A^\circ)}{P_B^\circ}$
15. For a zero-order reaction $\text{A} \rightarrow \text{B}$. At the end of 1 h, A is 75% reacted. How much of it will be left unreacted at the end of 2 h?
 (A) 12.5% (B) 6.25% (C) 3.12% (D) 0%
16. In general, the rate of a reaction can be increased by all the factors except
 (A) increasing the temperature (B) increasing the concentration of reactants
 (C) increasing the activation energy (D) using a catalyst
17. The only incorrect statement regarding Henry's law for a gas dissolving in water is
 (A) The value of K_H (Henry's constant) is greater for O_2 than H_2 at same temperature
 (B) The value of K_H increases with increase in temperature
 (C) Henry's law is not applicable for HCl(g)
 (D) Henry's law is applicable when the solubility of gas is low
18. The composition of a sample of cuprous oxide is found to be $\text{Cu}_{1.92}\text{O}_{1.00}$ due to metal deficiency defect. The molar ratio of Cu^{2+} and Cu^+ ions in the crystal is:
 (A) 4 : 5 (B) 1 : 12.5 (C) 1 : 23 (D) 1 : 24
19. The values of enthalpies of reactants and products are x and y J/mol, respectively. If the activation energy for the backward reaction is z J/mol, then the activation energy for forward reaction will be (in J/mol)
 (A) $x - y - z$ (B) $x - y + z$ (C) $y - x - z$ (D) $y - x + z$

20. For a reversible reaction $2\text{NO}_2 \xrightleftharpoons[k_2]{k_1} \text{N}_2\text{O}_4$, the rate of disappearance of NO_2 is equal to:
- (A) $\frac{2k_1}{k_2}[\text{NO}_2]^2$ (B) $2k_1[\text{NO}_2]^2 - 2k_2[\text{N}_2\text{O}_4]$
- (C) $2k_1[\text{NO}_2]^2 - k_2[\text{N}_2\text{O}_4]$ (D) $(2k_1 - k_2)[\text{NO}_2]$

SPACE FOR ROUGH WORK

SECTION-2

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21. X-ray diffraction shows that the unit cell edge length of NaCl is 500 pm. If the measured density of NaCl is 2.964 g/ml, what percent of Na^+ and Cl^- ions must be missing? ($N_A = 6 \times 10^{23}$)
22. Solutions A and B have osmotic pressures of 2.4 atm and 4.2 atm, respectively, at a certain temperature. The osmotic pressure (in atm) of a solution prepared by mixing the solutions in 2 : 1 volume ratio, respectively, at the same temperature is:
23. For the reaction $3\text{BrO}^- \rightarrow \text{BrO}_3^- + 2\text{Br}^-$ in alkaline aqueous solution, the value of the second-order rate constant (in BrO^-) in rate law for $-\frac{d[\text{BrO}^-]}{dt}$ was found to be $0.06 \text{ M}^{-1}\text{s}^{-1}$. The rate constant (in $\text{M}^{-1}\text{s}^{-1}$), when the rate law is written as $+\frac{d[\text{BrO}_3^-]}{dt}$ is 'a' $\text{M}^{-1}\text{s}^{-1}$ and as $+\frac{d[\text{Br}^-]}{dt}$ is 'b' $\text{M}^{-1}\text{s}^{-1}$, then the value of $(a + b) \times 100$ is:
24. The total number of ions per unit cell in sphalerite (zinc blend) structure if there is a Schottky defect of one anion per unit cell, is:
25. Consider two first-order reactions I and II. The frequency factor of I is 100 times that of II, the activation energy of I is 4.606 kcal higher than that of II. If the ratio of rate constants for the reactions I and II is $x : 1$ at 227°C , then the value of x is: [Given: $R = 2 \text{ cal/mol K}$, $e^{-4.606} = 0.01$]

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- $\int \frac{d(x^2+1)}{\sqrt{(x^2+2)}}$ is equal to

(A) $2\sqrt{(x^2+2)} + c$ (B) $\sqrt{(x^2+2)} + c$
 (C) $2x + 1$ (D) $2\sqrt{(x^2+1)} + c$
- $\int \left(\frac{\sec x}{\sec x - \tan x} \right) dx$ equal:

(A) $\sec x - \tan x + C$ (B) $\log|\sec x + \tan x| + C$
 (C) $\log|\sec x - \tan x| + C$ (D) $\sec x + \tan x + C$
- If the function $f(x) = \begin{cases} Ax^3 + B, & 0 \leq x \leq 1 \\ 2\cos(\pi x) + \tan^{-1} x, & 1 < x \leq 2 \end{cases}$ is differentiable in $[0, 2]$, then :

(A) $A + B = \frac{\pi - 8}{4}$ (B) $A - B = \frac{28 - 3\pi}{12}$ (C) $\frac{A}{B} = \frac{2}{3\pi - 26}$ (D) All of these
- Let $f(x) = \log_e \left\{ \frac{u(x)}{v(x)} \right\}$, $u'(2) = 4$, $v'(2) = 2$, $u(2) = 2$, $v(2) = 1$, then $f'(2)$ is equal to :

(A) 0 (B) 1 (C) -1 (D) None of these
- Let the tangent at $(1, 1)$ on the curve $y^2 = x(2-x)^2$ meet the curve again at point P . If the coordinate of P is (α, β) , then $(4\alpha + 16\beta)$ equals :

(A) 15 (B) 12 (C) 10 (D) 9
- If $\int \sec^2 x \operatorname{cosec}^4 x \, dx = f(x)$ and $f\left(\frac{\pi}{6}\right) = -\frac{8}{\sqrt{3}}$, then $f\left(\frac{\pi}{4}\right)$ is :

(A) 1 (B) $-\frac{1}{3}$ (C) $-\frac{4}{3}$ (D) 0
- The range of ' α ' for which all the points of local extrema of the function $f(x) = x^3 - 3\alpha x^2 + 3(\alpha^2 - 1)x + 1$ lie in the interval $(-2, 4)$, is :

(A) $(-1, 3)$ (B) $(3, 4)$ (C) $(-4, -2)$ (D) $(-2, -1)$
- If $f(x) = \begin{cases} (x+1)2^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then :

(A) $f(x)$ is continuous at $x = 0$ (B) $f(x)$ is differentiable at $x = 0$
 (C) $\lim_{x \rightarrow 0^+} f(x)$ does not exist (D) $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$

9. Let the function $f(x)$ be defined as

$$f(x) = \begin{cases} \frac{\ln x - 1}{x - e}, & x \neq e \\ k, & x = e \end{cases}$$

The value of k , for which the function is continuous at $x = e$, is equal to:

- (A) e (B) $\frac{1}{e}$ (C) e^2 (D) $-e$
10. Let $f(x) = |x| \sin x + |x^2 - \pi^2| \cos x$, then the number of points where $y = f(x)$ is NOT differentiable is/are:
- (A) 0 (B) 1 (C) 2 (D) 3

11. If all the real values of m for which the function $f(x) = \frac{x^3}{3} - (m-3)\frac{x^2}{2} + mx - 2020$ is strictly increasing for $x \in R$ is $[a, b]$ then $a + b$ is :

- (A) 9 (B) 10 (C) 11 (D) 12

12. If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx$ & $f(0) = 0$, then the value of $f(1)$ is :

- (A) $-\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $-\frac{1}{4}$

13. The equation $e^{x-1} + x - 2 = 0$ has:

- (A) One real root (B) Two real roots
(C) Three real roots (D) Four real roots

14. $\int x \frac{e^{\tan^{-1} x^2}}{x^4 + 1} dx$ equals :

- (A) $e^{x^2} \tan^{-1} x^2 + c$ (B) $\frac{1}{2} e^{\tan^{-1} x^2} + c$
(C) $\log(\tan^{-1} x^2) + c$ (D) $\frac{x^2}{2} \log(x^2) + c$

15. If $f(x) = \lim_{t \rightarrow 0} \tan^{-1} \left(\frac{e^{xt} - 1}{t} \right)$, then the value of $\lim_{x \rightarrow 0} \frac{f(x) - x}{x^3}$ is equal to :

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

16. If $y = \frac{x^{12} + x^6 + 1}{x^6 + x^3 + 1}$ & $\frac{dy}{dx} = x^2(ax^3 + b)$, then the value of $(a + b)$ is equal to :

- (A) 2 (B) 3 (C) 5 (D) 9

17. Let $f''(x) > 0 \forall x \in R$ and let $g(x) = f(x) + f(2-x)$, then interval of x for which $g(x)$ is increasing is:

- (A) $(1, \infty)$ (B) R (C) $[-1, 1]$ (D) $[-2, \infty)$

18. If $\lim_{n \rightarrow \infty} \frac{e\left(1 - \frac{1}{n}\right)^n - 1}{n^\alpha}$, exists and is equal to l ($l \neq 0$), then the value of $12(l - \alpha)$ is :

- (A) 4 (B) 3 (C) 6 (D) 7

19. $\int \frac{a^{\sqrt{x}}}{\sqrt{x}} dx$ equals;
- (A) $2a^{\sqrt{x}} \log_e a + c$ (B) $2a^{\sqrt{x}} \log_a e + c$
 (C) $2a^{\sqrt{x}} \log_{10} a + c$ (D) $2a^{\sqrt{x}} \log_a 10 + c$
20. The radius of cylinder of maximum volume which can be inscribed in a right circular cone of radius R and height H (axis of cylinder and cone are same) given by :
- (A) $\frac{R}{2}$ (B) $\frac{R}{3}$ (C) $\frac{2R}{3}$ (D) $\frac{2R}{5}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08)

21. The number of points where $g(x) = \min\{|x-3|, 9-x^2\}$ ($x \in R$) is not differentiable is _____.
22. If $f(x) = f(1-x)$ and $f(x)$ differentiable at every real value of x , then the value of $f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right)$ is _____.
23. The integer 'n' for which the value of $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(e^x - \cos x) - \frac{x^3}{2}}{x^n}$ is a finite non zero number, is _____.
24. If f is a differentiable function for all real x and $f'(x) \leq 5 \forall x \in R$. If $f(2) = 0$ and $f(5) = 15$ value of $\frac{f(3)}{2}$ is _____.
25. If $f(x) = \begin{cases} [x] & ; 0 \leq x < 3 \\ |[x]+1| & ; -3 < x < 0 \end{cases}$, where $[.]$ denotes greatest integer function, then number of point at which $f(x)$ is discontinuous in $(-3, 3)$, is equal to _____.

SPACE FOR ROUGH WORK

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